ML Lab Week 10 SVM Lab

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SRN: PES2UG23CS160

MOON DATASET:

```
SVM with LINEAR Kernel <PES2UG23CS160>
              precision    recall  f1-score   support

           0       0.85      0.89      0.87        75
           1       0.89      0.84      0.86        75

    accuracy                           0.87       150
   macro avg       0.87      0.87      0.87       150
weighted avg       0.87      0.87      0.87       150

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SVM with RBF Kernel <PES2UG23CS160>
              precision    recall  f1-score   support

           0       0.95      1.00      0.97        75
           1       1.00      0.95      0.97        75

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150

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SVM with POLY Kernel <PES2UG23CS160>
...
weighted avg       0.86      0.85      0.85       150
```

Moons Dataset Questions (2 questions):

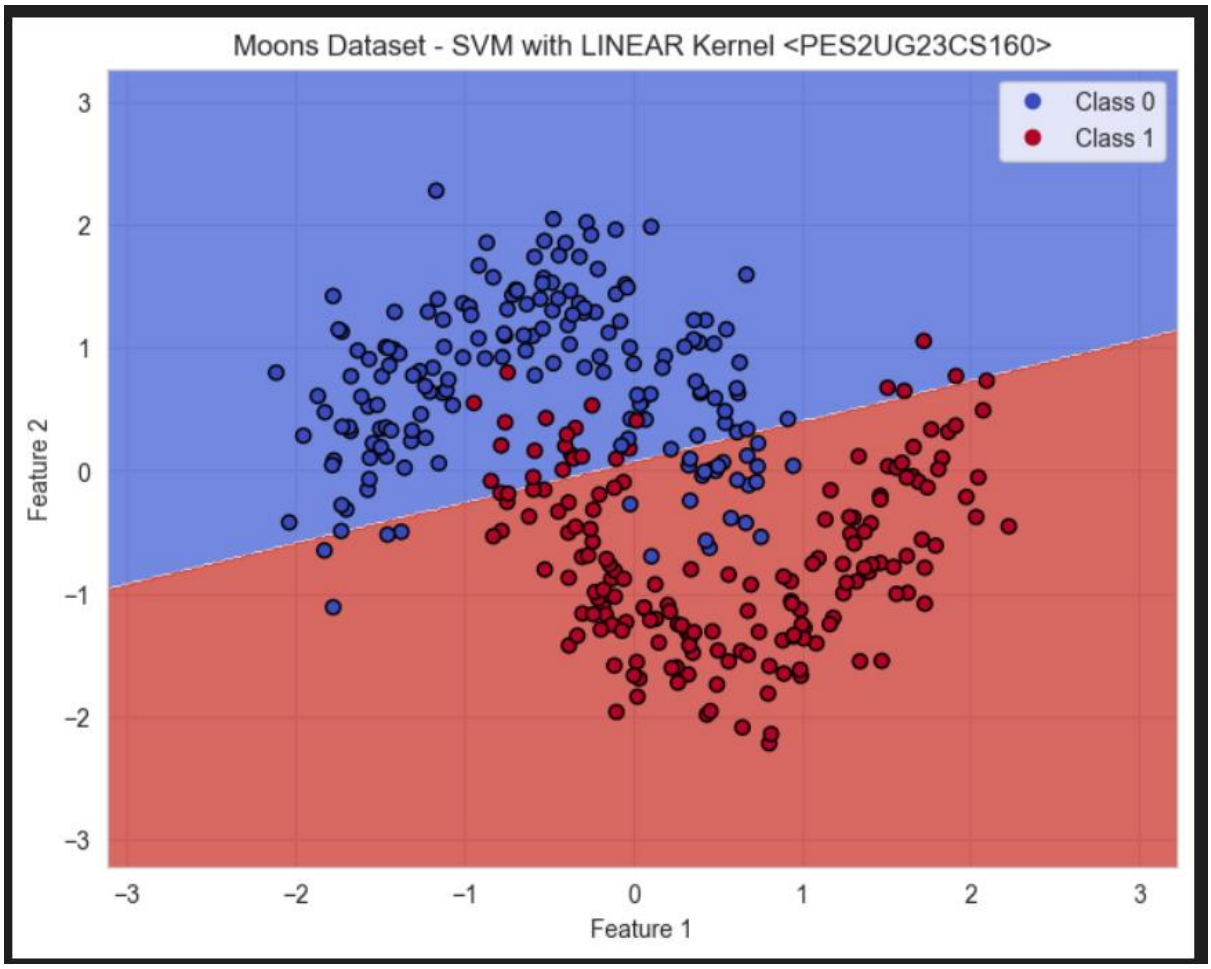
1. **Inferences about the Linear Kernel's performance.**

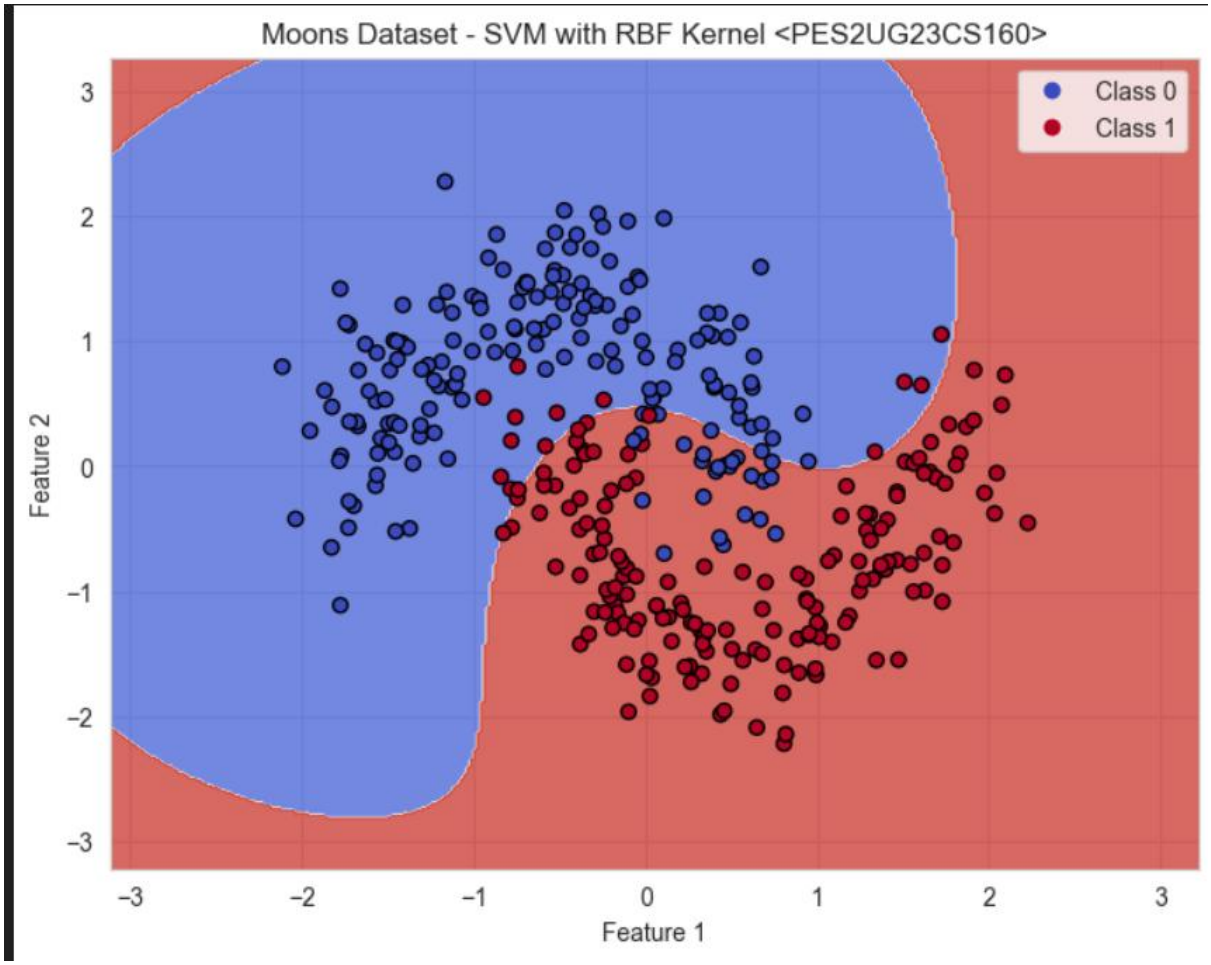
An accuracy of **87%** shows that the model performs significantly better than random guessing, but it's far from perfect. It establishes a performance **baseline**, but the 13% error rate indicates a fundamental mismatch between the model and the data.

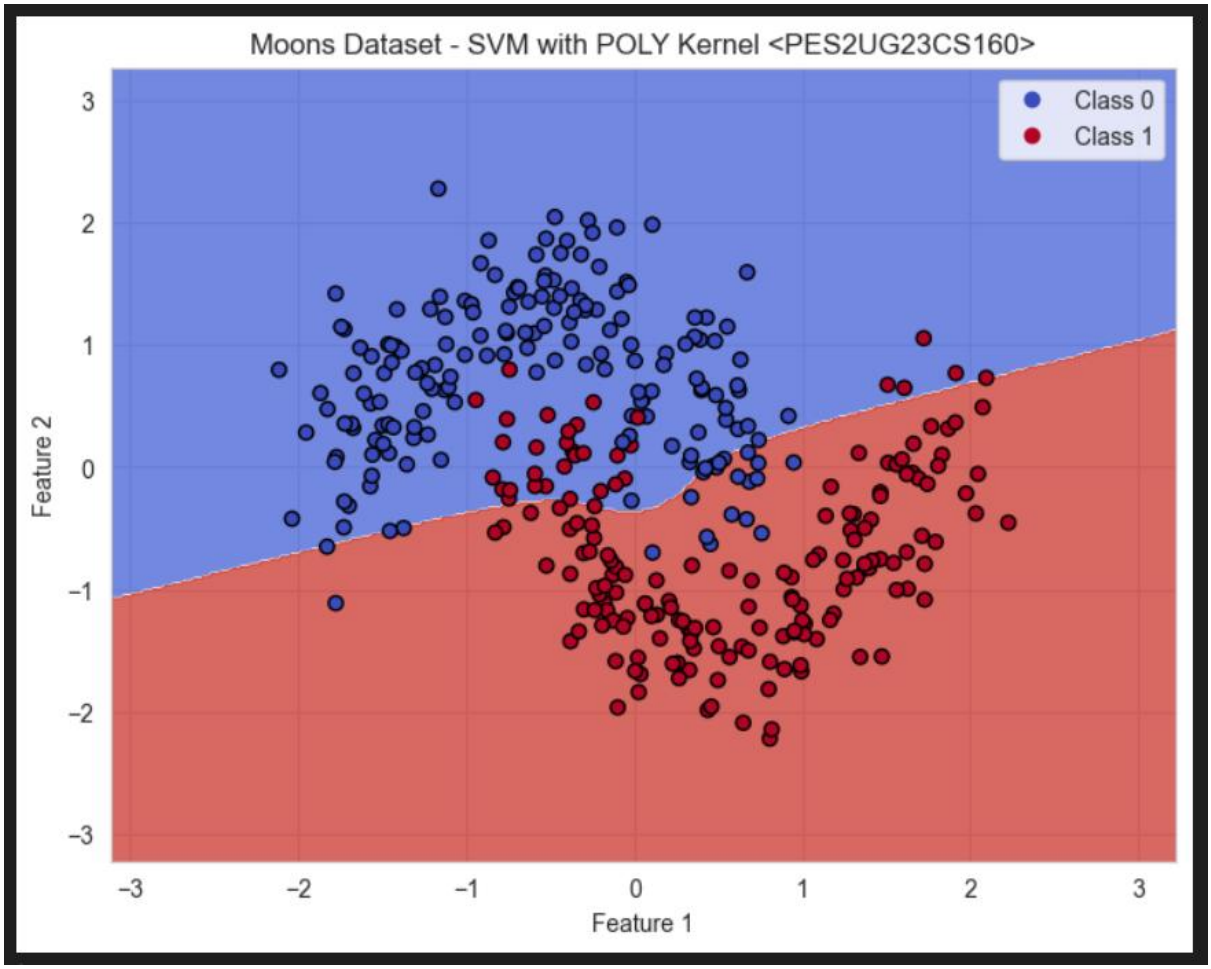
2. **Comparison between RBF and Polynomial kernel decision boundaries.**

The RBF kernel produced a smooth, non-linear curve that accurately follows the crescent shape of the two moons. This boundary respects the underlying structure of the data, leading to a high accuracy of 97%.

The Polynomial kernel (with its default settings) produced a boundary that was almost a straight line, very similar to the Linear kernel. It failed to create the necessary curve to separate the classes effectively, resulting in poor performance.




Moons Dataset - SVM with RBF Kernel <PES2UG23CS160>
Class 0
Class 1
Feature 2
Feature 1
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3



BANK DATASET:

```
SVM with LINEAR Kernel <PES2UG23CS160>
              precision    recall  f1-score   support

      Forged       0.90      0.88      0.89       229
     Genuine       0.86      0.88      0.87       183

    accuracy                           0.88       412
   macro avg       0.88      0.88      0.88       412
weighted avg       0.88      0.88      0.88       412

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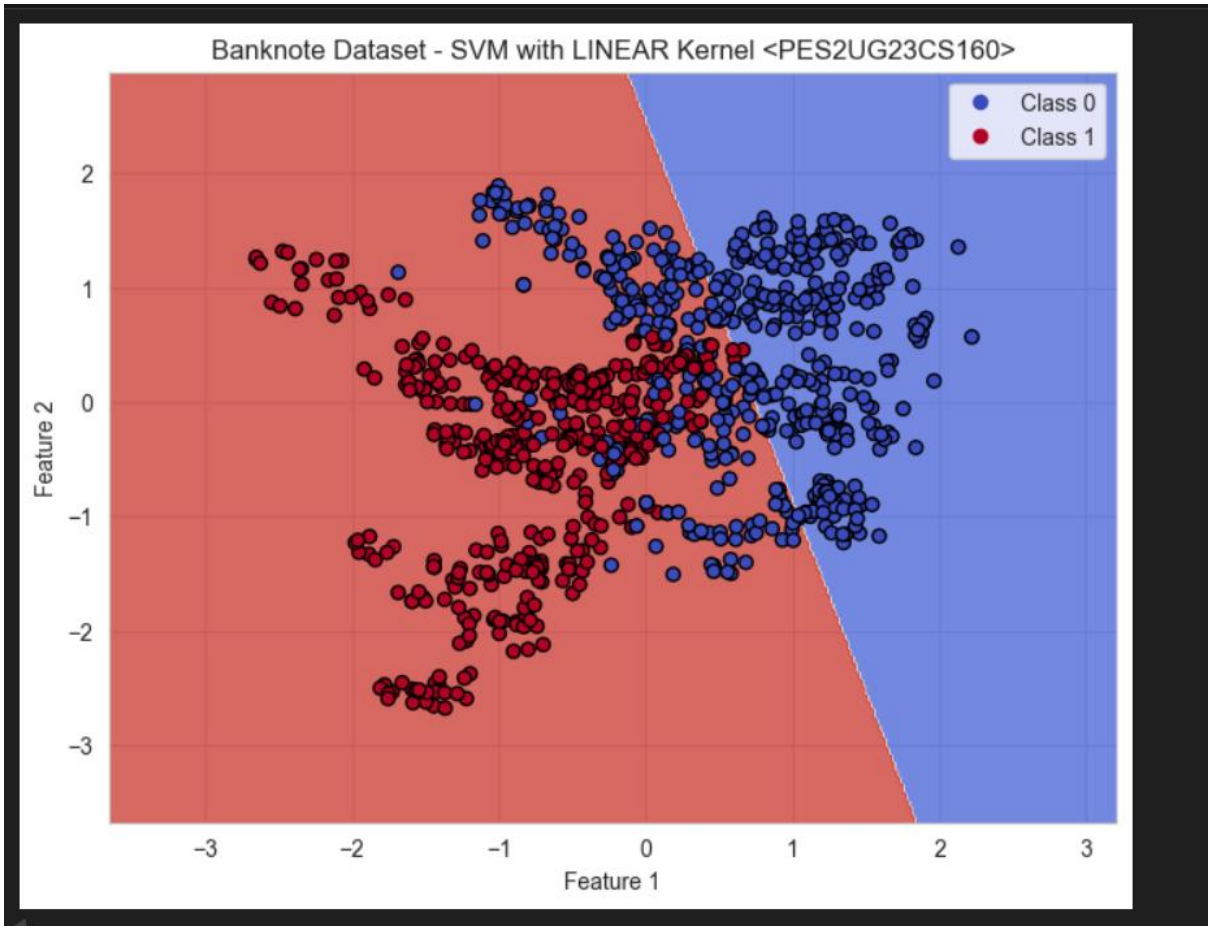
SVM with RBF Kernel <PES2UG23CS160>
              precision    recall  f1-score   support

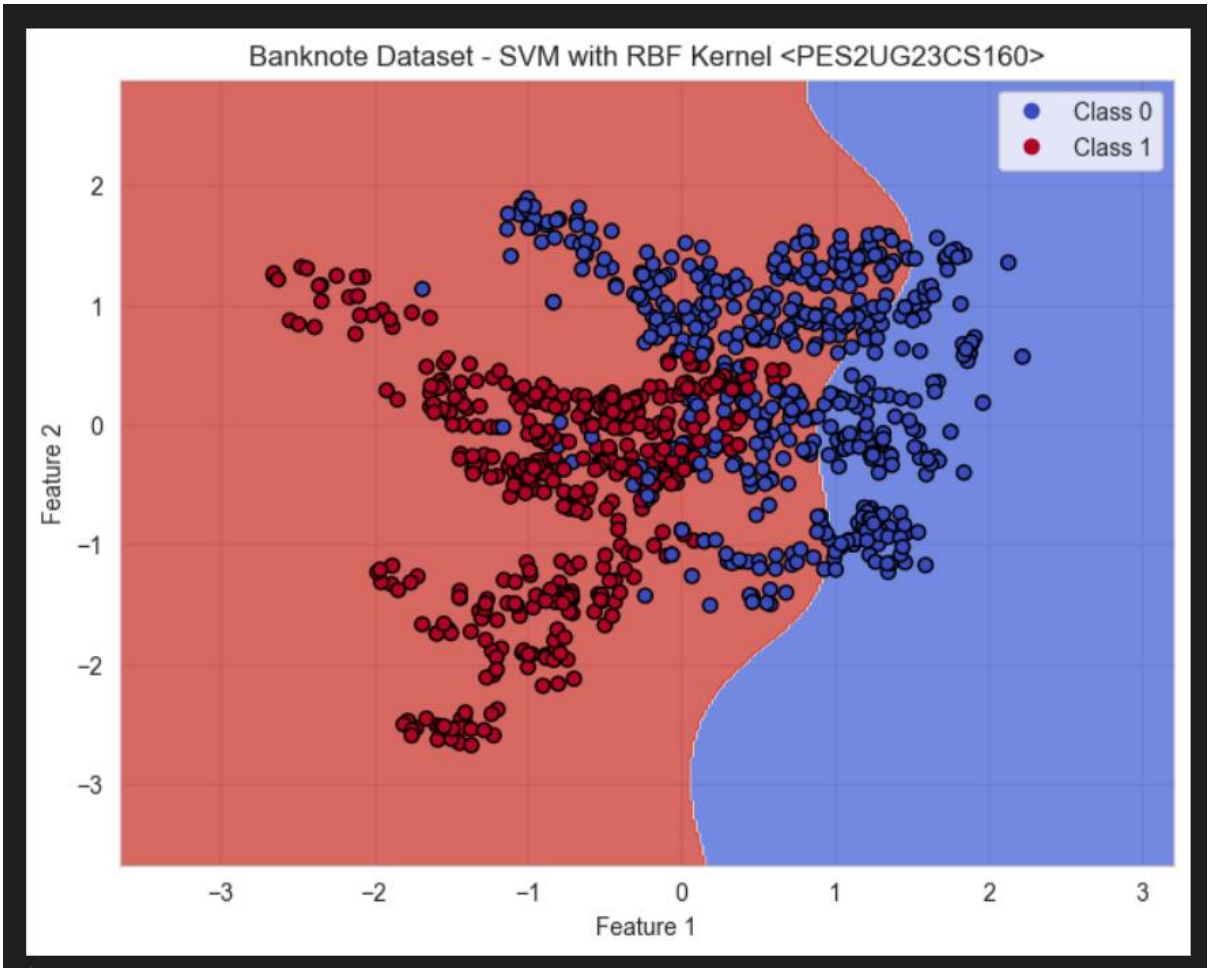
      Forged       0.98      0.90      0.94       229
     Genuine       0.89      0.98      0.93       183

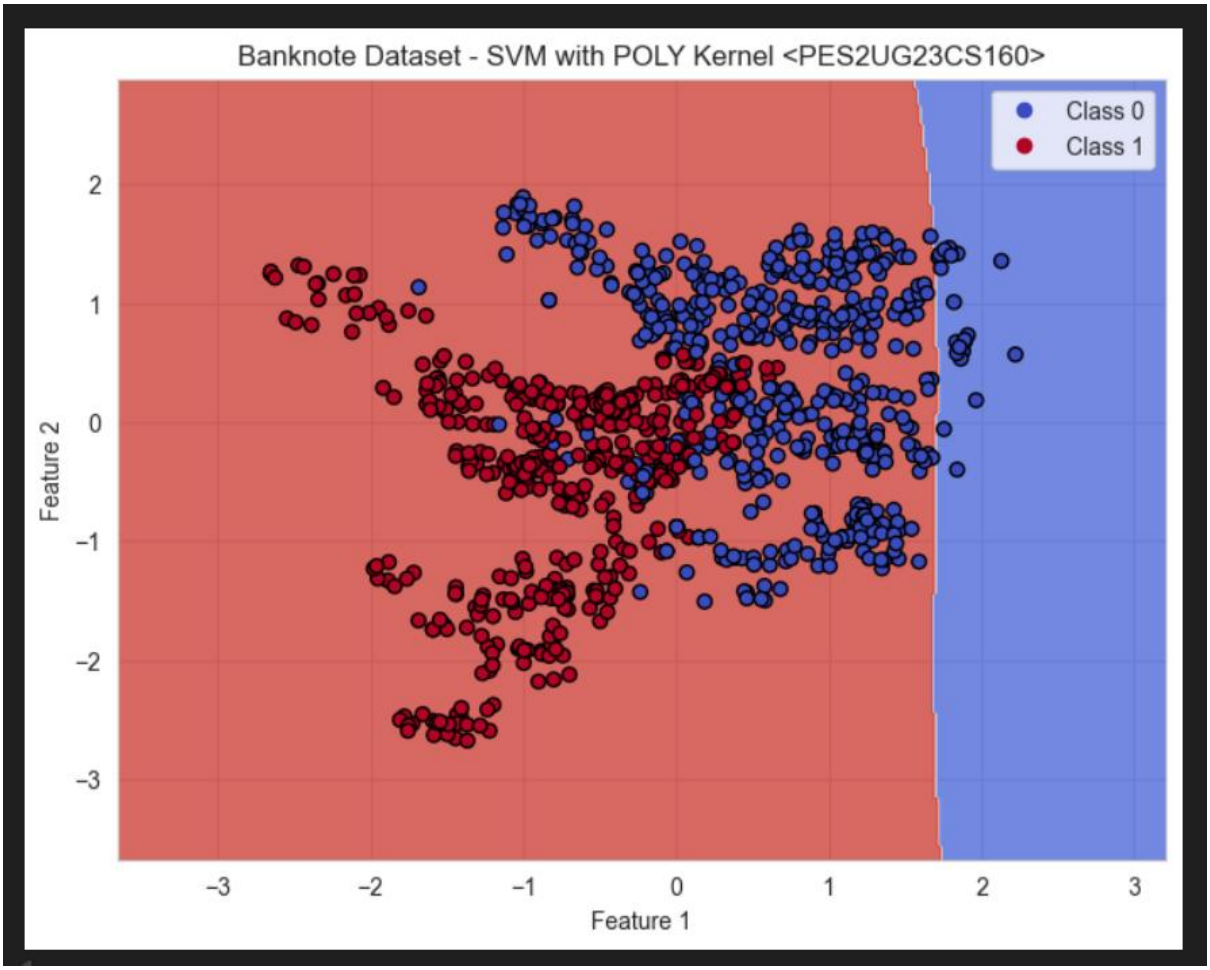
    accuracy                           0.94       412
   macro avg       0.94      0.94      0.94       412
weighted avg       0.94      0.94      0.94       412

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SVM with POLY Kernel <PES2UG23CS160>
...
weighted avg       0.91      0.90      0.90       412
```


Banknote Dataset - SVM with LINEAR Kernel <PES2UG23CS160>
Class 0
Class 1
Feature 2
Feature 1
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

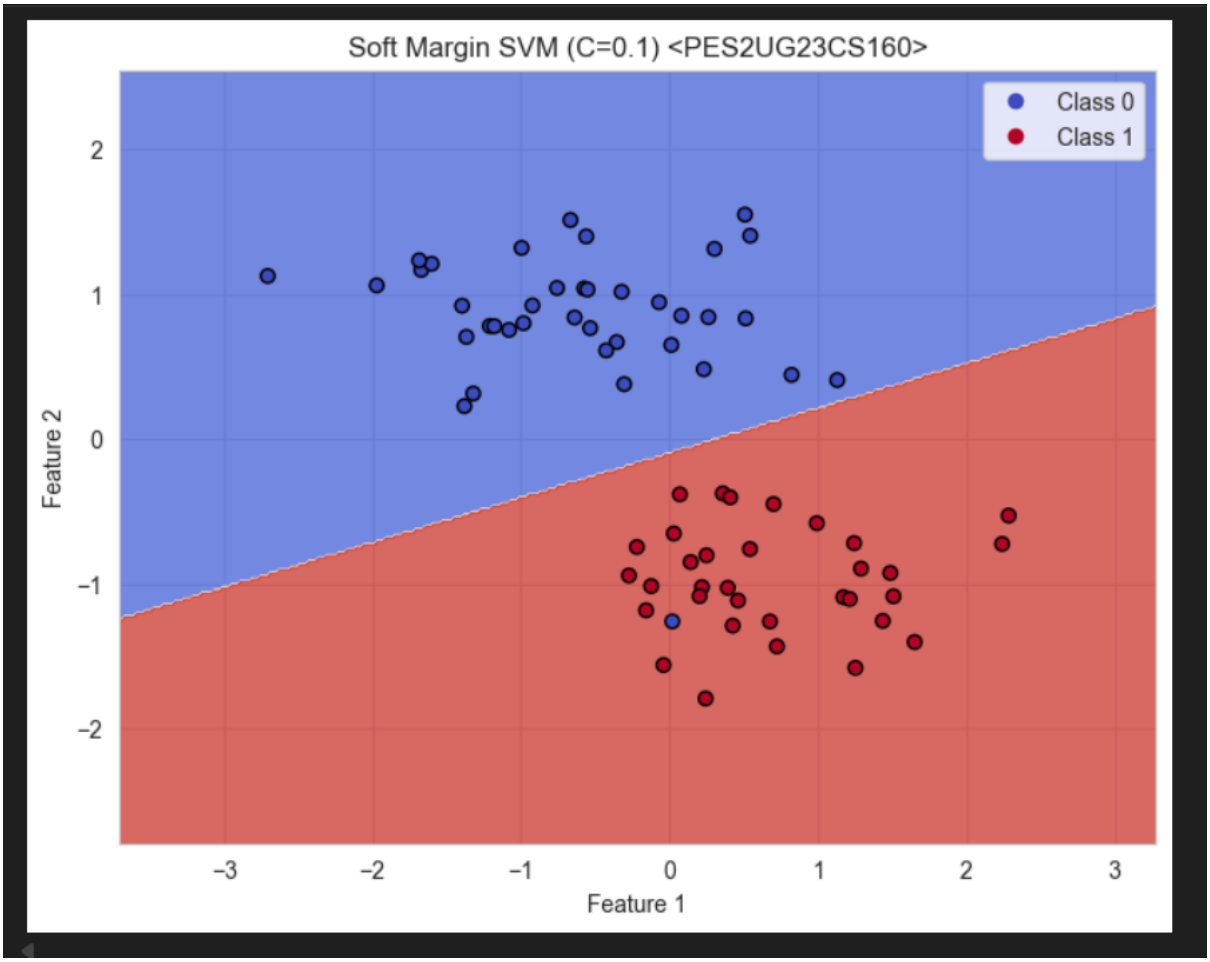

Banknote Dataset - SVM with RBF Kernel <PES2UG23CS160>
Class 0
Class 1
Feature 2
Feature 1
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

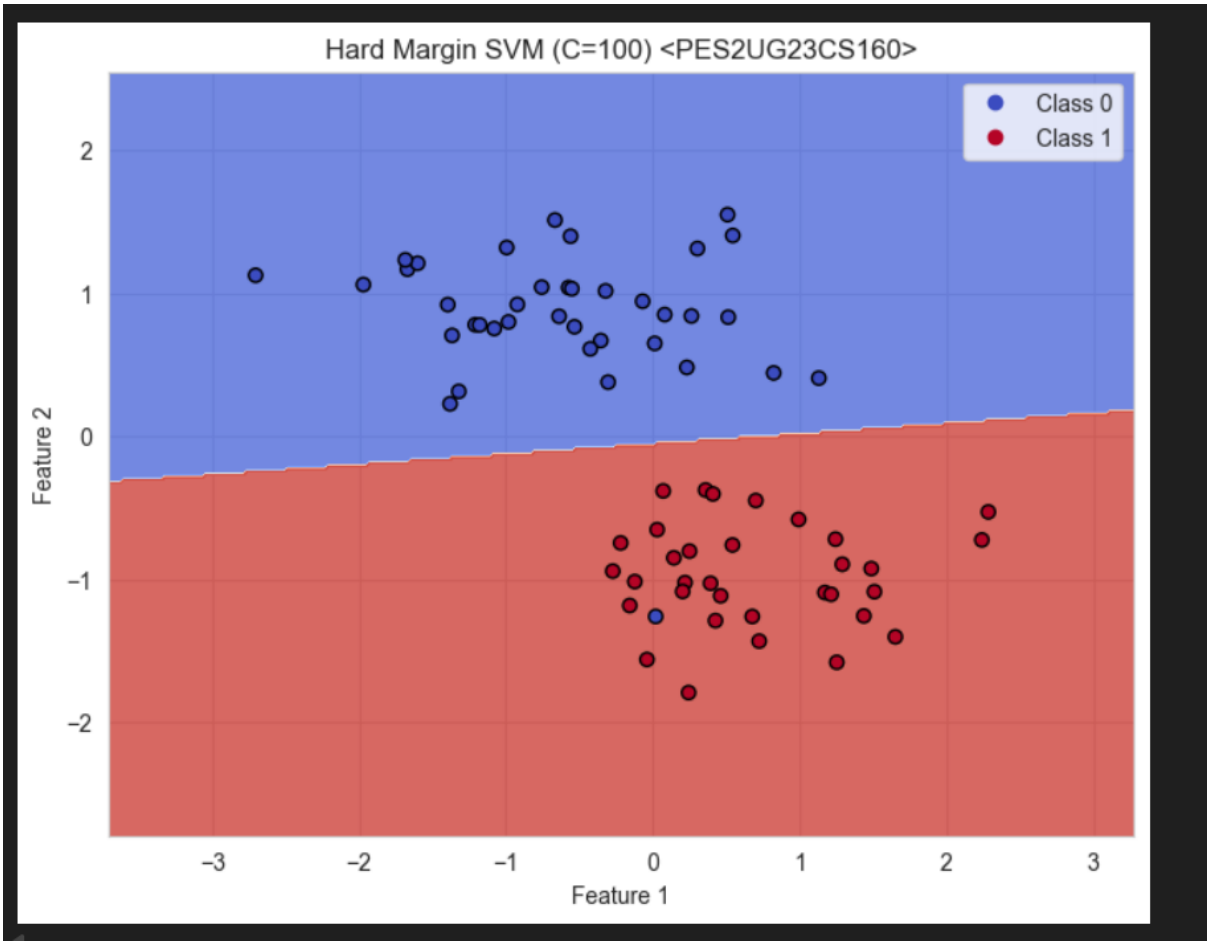


Banknote Dataset Questions (2 questions):

1. **Which kernel was most effective for this dataset?**
   It achieved the highest accuracy (**94%**) compared to the Linear (88%) and Polynomial (~91%) kernels. The RBF kernel's flexible, non-linear decision boundary was best able to model the complex and overlapping distribution of the genuine and forged banknote data.
2. **Why might the Polynomial kernel have underperformed here?**
   The Polynomial kernel likely underperformed on the Banknote dataset for two primary reasons: a mismatch with the data's structure and a lack of hyperparameter tuning.

   MARGIN ANALYSIS:


Soft Margin SVM (C=0.1) <PES2UG23CS160>
Class 0
Class 1
Feature 2
Feature 1
2
1
0
−1
−2
−3
−2
−1
0
1
2
3



Hard vs. Soft Margin Questions (4 questions):

1. **Which margin (soft or hard) is wider?**
   The "Soft Margin" (*C*=0.1) model produces a noticeably wider margin.

2. **Why does the soft margin model allow "mistakes"?**

   The SVM allows these "mistakes" because of the trade-off between margin width and classification error. With a low *C* value (0.1), the penalty for misclassifying a training point is small. Therefore, the model's optimizer chooses to ignore the outlier (the blue dot in the red cluster) because doing so allows it to create a much wider and simpler margin for the rest of the data.

3. **Which model is more likely to be overfitting and why?**
   The "Hard Margin" (*C*=100) model is much more likely to be overfitting.
4. **Which model would you trust more for new data and why?**

   For a new, unseen data point, you should trust the "Soft Margin" (*C*=0.1) model more. Because it is less likely to be overfitted. Its boundary reflects the general pattern of the data, not the peculiarities of a single outlier. This makes it more robust and more likely to classify a new data point correctly